ZEPHYR A. FLANIGAN

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EDUCATION

The George Washington University – Washington, DC *Sep 2021 - Jan 2025*
Political Science, minor Computer Science;

EXPERIENCE

EnergyCAP – Remote
Associate Software Engineer *Jan 2026 – Present*

- Architected and delivered full-stack capabilities for an AI-powered document processing platform using C#/.NET 10, SQL Server, Razor, and TypeScript, spanning REST API's, database migrations, backend workflows, and frontend UI.
- Designed and maintained core document ingestion workflow, including bulk file uploads, status polling, reprocessing endpoints, EF core migrations with data preservation, and structured routing of extracted bill data into application workflows.
- Built reliability, validation, and testing infrastructure, including field-level acceptance testing with fuzzy string matching, input normalization, and 24+ unit and integration tests using xUnit, Moq, and Testcontainers.

Data Analyst Intern *May 2025 – Dec 2025*

- Created web apps using aggregated company API data to improve workflow automation for 20 person Managed Services Team.
- Built and maintained Power BI dashboards tracking critical KPI data, supporting data-driven roadmap decisions for engineering and product leadership.
- Developed SLA reports to identify partner violations resulting in over $35,000 in credits within the first month of implementation.
- Cleaned, normalized, and validated unstructured data for over 13,000 vendors by writing advanced SQL queries and developing Python scripts using pandas and numpy, improving data quality and reliability across the company's platform.

Federal Communications Commission – Washington, DC
National Security Intern *June 2024 – Sep 2024*

- Analyzed cybersecurity regulations from multiple agencies to evaluate where terminology could be standardized.
- Developed a baseline taxonomy of language to be used as guidance for future regulations to ensure homogenization.
- Implemented the Cybersecurity Regulatory Forum's model regulatory language product using the NIST Cybersecurity Framework 2.0 to minimize financial burden on regulated entities.

The George Washington University Campus Living – Washington, DC
Community Engagement Assistant *August 2023 – May 2024*

- Conducted analysis of community trends through surveys to curate and update community bulletin boards bi-monthly, ensuring relevance and engagement.
- Organized multiple programs and events aimed at fostering interpersonal relationships within the community.
- Engaged proactively with residents to assess their needs and strategically referred them to the appropriate campus resources.

PROFESSIONAL PROJECTS

In-house Bill Extraction Software

- Co-developed an internal AI/LLM document processing system to extract structured data from utility bill PDFs for integration with the company's core software platform.
- Trained and evaluated LLM-powered extraction workflow on hundreds of utility bill documents sourced from internal production databases, improving support for varied bill formats, vendors, and layouts.
- Reduced bill extraction costs by over 50% and improved turnaround from an average of 2 days to 15 minutes per bill by reducing dependency on external extraction workflows.

Kickout Manager Dashboard

- Developed and deployed a full-stack web application using Python and ReactJS that integrated with the company API to monitor and track billing exceptions (kickouts) in real-time, updating hourly.
- Aggregated and visualized kickout data for 300+ customers, enabling correction of 15,000+ billing errors.
- Reduced Managed Services team's kickout resolution time by 50% through automated tracking and providing centralized visibility into billing exceptions.

ADDITIONAL INFORMATION AND SKILLS

Programming Languages: C#, Bash, HTML/CSS, Java, Python, SQL, TypeScript, Angular, React
Platform/Tools: SSMS, Docker, PySpark, Hadoop, Microsoft Office, Pandas/NumPy/Seaborn, PgAdmin, PowerBI, Tableau, Azure